

# INDETERMINATE FORM — COMPLETE SOLVED NOTES

Extracted from your question sheet • Easy methods + alternative methods • Exam-focused

**Indeterminate forms:  $0/0$ ,  $\infty/\infty$ ,  $0 \cdot \infty$ ,  $\infty - \infty$ ,  $0^0$ ,  $1^\infty$  and  $\infty^0$ . First substitute, identify the form, then choose the shortest valid method.**

## EXTRACTED QUESTIONS

1.  $\lim_{x \rightarrow 0} [x e^x - \ln(1+x)] / x^2$
2.  $\lim_{x \rightarrow 0} [e^x - e^{-x} + 2\sin x - 4x] / x^5$
3.  $\lim_{\theta \rightarrow \pi/4} [\sqrt{2} - \cos\theta - \sin\theta] / (4\theta - \pi)^2$
4.  $\lim_{x \rightarrow a} (x^n - a^n) / (x - a)$
5.  $\lim_{x \rightarrow 3} (x^2 - 9) / (x - 3)$
6.  $\lim_{x \rightarrow 0} (\sqrt[3]{1+x} - 1) / x$
7.  $\lim_{x \rightarrow a} [x^{7/2} - a^{7/2}] / (\sqrt{x} - \sqrt{a})$
8.  $\lim_{x \rightarrow 0} (1 - \cos x) / x^2$
9.  $\lim_{x \rightarrow 0} x^2 \log(x^2)$
10.  $\lim_{x \rightarrow 0} (\tan x - \sin x) / x^3$
11. Evaluate using L'Hôpital:  $\lim_{x \rightarrow 0^+} (1/x)^{(\tan x)}$
12. Evaluate using L'Hôpital:  $\lim_{x \rightarrow \infty} (e^x + 1)^{(-2/x)}$

**Exam workflow: Substitute → identify indeterminate form → try factorization/rationalization/trig identity → use L'Hôpital for  $0/0$  or  $\infty/\infty$  → for power forms take  $\ln$  first.**

### Problem 1

Evaluate  $\lim_{x \rightarrow 0} (x e^x - \ln(1+x))/x^2$

*Analysis: Substitution gives 0/0, so L'Hôpital applies. Two applications are enough.*

#### **Method 1 — L'Hôpital:**

$$L = \lim_{x \rightarrow 0} [x e^x - \ln(1+x)]/x^2$$

$$\text{First differentiation: } L = \lim_{x \rightarrow 0} [(1+x)e^x - 1/(1+x)]/(2x)$$

*Still 0/0. Differentiate again:*

$$L = \lim_{x \rightarrow 0} [(2+x)e^x + 1/(1+x)^2] / 2$$

$$= [2 + 1]/2 = 3/2.$$

**$\therefore$  Answer:  $L = 3/2$**

*Method 2 — Series check:  $x e^x = x + x^2 + O(x^3)$ , and  $\ln(1+x) = x - x^2/2 + O(x^3)$ . Thus numerator =  $3x^2/2 + O(x^3)$ , giving  $3/2$ .*

## Problem 2

Evaluate  $\lim_{x \rightarrow 0} (e^x - e^{-x} + 2\sin x - 4x)/x^5$

*Analysis: Direct substitution gives 0/0. Lower-order terms cancel, so Taylor series is much shorter than five rounds of L'Hôpital.*

### **Method 1 — Taylor series (easiest):**

$$e^x - e^{-x} = 2x + x^3/3 + x^5/60 + O(x^7)$$

$$2\sin x = 2x - x^3/3 + x^5/60 + O(x^7)$$

$$\begin{aligned}\text{Numerator} &= (2x - 4x + 2x) + (x^3/3 - x^3/3) + (x^5/60 + x^5/60) + O(x^7) \\ &= x^5/30 + O(x^7).\end{aligned}$$

$$\text{Therefore } L = 1/30.$$

**$\therefore$  Answer:  $L = 1/30$**

*Method 2 — repeated L'Hôpital: Five differentiations reduce the problem to the fifth derivative at 0 divided by 5!. The fifth-order coefficient is 1/30, so the same answer is obtained. The series method is cleaner here.*

### Problem 3

Evaluate  $\lim_{\theta \rightarrow \pi/4} [\sqrt{2} - \cos\theta - \sin\theta]/(4\theta - \pi)^2$

*Analysis: Substitution gives 0/0. Use the identity  $\cos\theta + \sin\theta = \sqrt{2} \cos(\theta - \pi/4)$ .*

#### **Method 1 — Trig identity (easiest):**

Numerator =  $\sqrt{2}[1 - \cos(\theta - \pi/4)]$ . Let  $h = \theta - \pi/4$ , so  $4\theta - \pi = 4h$ .

$$L = \lim_{h \rightarrow 0} \sqrt{2}(1 - \cosh)/(16h^2)$$

$$= (\sqrt{2}/16) \cdot \lim_{h \rightarrow 0} (1 - \cosh)/h^2 = (\sqrt{2}/16)(1/2).$$

**$\therefore$  Answer:  $L = \sqrt{2}/32$**

*Method 2 — L'Hôpital twice: Differentiate numerator and denominator twice. At  $\theta = \pi/4$  the second derivative of the numerator is  $\sqrt{2}$  and that of the denominator is 32, giving  $\sqrt{2}/32$ .*

#### Problem 4

Evaluate  $\lim_{x \rightarrow a} (x^n - a^n)/(x - a)$

*Analysis: The form is 0/0. Factorization is the cleanest method.*

##### **Method 1 — Difference of powers:**

$$x^n - a^n = (x - a)(x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + a^{n-1})$$

Cancel  $x - a$ , then put  $x = a$ . There are  $n$  equal terms  $a^{n-1}$ .

$$L = n a^{n-1}.$$

$$\therefore \text{Answer: } L = n a^{n-1}$$

Method 2 — L'Hôpital:  $L = \lim_{x \rightarrow a} [n x^{n-1}]/1 = n a^{n-1}$ .

### Problem 5

Evaluate  $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$

*Analysis: Form 0/0. Factorization is immediate.*

#### **Method 1 — Factorization:**

$$x^2 - 9 = (x - 3)(x + 3)$$

$$L = \lim_{x \rightarrow 3} (x + 3) = 6.$$

**$\therefore$  Answer:  $L = 6$**

Method 2 — L'Hôpital:  $L = \lim_{x \rightarrow 3} (2x) = 6.$

### Problem 6

Evaluate  $\lim_{x \rightarrow 0} [\sqrt{1+x}-1]/x$

*Analysis: Form 0/0. Rationalization is the shortest method.*

#### **Method 1 — Rationalization:**

$$\begin{aligned} L &= \lim_{x \rightarrow 0} [(\sqrt{1+x}-1)(\sqrt{1+x}+1)]/[x(\sqrt{1+x}+1)] \\ &= \lim_{x \rightarrow 0} [x]/[x(\sqrt{1+x}+1)] \\ &= 1/(\sqrt{1+x}+1) \rightarrow 1/2. \end{aligned}$$

**$\therefore$  Answer:  $L = 1/2$**

Method 2 — L'Hôpital:  $[1/(2\sqrt{1+x})]/1$  at  $x=0 = 1/2$ .

### Problem 7

Evaluate  $\lim_{x \rightarrow a} [x^{7/2} - a^{7/2}] / (\sqrt{x} - \sqrt{a})$

*Analysis: Form 0/0. Put  $t = \sqrt{x}$  so the numerator becomes  $t^7$ .*

#### **Method 1 — Substitution + difference of powers:**

Let  $t = \sqrt{x}$ . Then  $x^{7/2} = t^7$  and  $t \rightarrow \sqrt{a}$ .

$$L = \lim_{t \rightarrow \sqrt{a}} [t^7 - (\sqrt{a})^7] / [t - \sqrt{a}]$$

$$= 7(\sqrt{a})^6 = 7a^3.$$

$\therefore \text{Answer: } L = 7a^3$

Method 2 — L'Hôpital: Numerator derivative =  $(7/2)x^{5/2}$ , denominator derivative =  $1/(2\sqrt{x})$ . Ratio =  $7x^3 \rightarrow 7a^3$ .

### Problem 8

Evaluate  $\lim_{x \rightarrow 0} (1 - \cos x)/x^2$

*Analysis: Form 0/0. Use the half-angle identity.*

#### **Method 1 — Identity:**

$$1 - \cos x = 2\sin^2(x/2)$$

$$L = 2\sin^2(x/2)/x^2 = 1/2 \cdot [\sin(x/2)/(x/2)]^2.$$

Using  $\sin u/u \rightarrow 1$ ,  $L = 1/2$ .

**$\therefore$  Answer:  $L = 1/2$**

Method 2 — L'Hôpital:  $L = \lim_{x \rightarrow 0} \sin x/(2x) = 1/2$ .

### Problem 9

Evaluate  $\lim_{x \rightarrow 0} x^2 \log(x^2)$

*Analysis:*  $x^2 \rightarrow 0^+$  and  $\log(x^2) \rightarrow -\infty$ , so the form is  $0 \cdot (-\infty)$ . Rewrite as a quotient.

#### **Method 1 — L'Hôpital:**

$$L = \lim_{x \rightarrow 0^+} \ln(x^2)/(1/x^2)$$

$$= \lim_{x \rightarrow 0^+} [2/x]/[-2/x^3] = \lim_{x \rightarrow 0^+} (-x^2) = 0.$$

**$\therefore$  Answer:  $L = 0$**

Method 2 — Substitution: Put  $y=x^2$ . Then  $y \rightarrow 0^+$  and  $y \ln y \rightarrow 0$ .

### Problem 10

Evaluate  $\lim_{x \rightarrow 0} (\tan x - \sin x)/x^3$

*Analysis: Form 0/0. A trig identity gives the answer in three standard limits.*

#### **Method 1 — Trig identity (easiest):**

$$\tan x - \sin x = \sin x(1/\cos x - 1) = \sin x(1 - \cos x)/\cos x.$$

$$\text{So } L = [\sin x/x] \cdot [(1 - \cos x)/x^2] \cdot [1/\cos x].$$

$$\text{As } x \rightarrow 0: 1 \cdot 1/2 \cdot 1 = 1/2.$$

**$\therefore$  Answer:  $L = 1/2$**

*Method 2 — L'Hôpital: It can be solved by repeated differentiation, but the identity method is considerably shorter and less error-prone.*

### Problem 11

Evaluate using L'Hôpital:  $\lim_{x \rightarrow 0^+} (1/x)^{\tan x}$

*Analysis: This is  $\infty^0$ . For an indeterminate power, take  $\ln$  first.*

#### **Method — Logarithm + L'Hôpital:**

Let  $L = \lim_{x \rightarrow 0^+} (1/x)^{\tan x}$ .

$\ln L = \lim_{x \rightarrow 0^+} \tan x \cdot \ln(1/x) = \lim_{x \rightarrow 0^+} [\ln(1/x)]/\cot x$ .

This is  $\infty/\infty$ . L'Hôpital gives  $\ln L = \lim_{x \rightarrow 0^+} [-1/x]/[-\csc^2 x] = \lim_{x \rightarrow 0^+} \sin^2 x/x$ .

Since  $\sin x \sim x$ ,  $\sin^2 x/x \sim x \rightarrow 0$ . Hence  $\ln L = 0$ .

Therefore  $L = e^0 = 1$ .

**$\therefore$  Answer:  $L = 1$**

### Problem 12

Evaluate using L'Hôpital:  $\lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$

*Analysis: This is  $\infty^0$ . Take natural logarithms.*

#### **Method — Logarithm + L'Hôpital:**

Let  $L = \lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$ .

$\ln L = \lim_{x \rightarrow \infty} [-2 \ln(e^x + 1)]/x$ .

This is  $(-\infty)/\infty$ . Apply L'Hôpital:

$\ln L = \lim_{x \rightarrow \infty} [-2e^x/(e^x + 1)] = -2$ .

Therefore  $L = e^{-2}$ .

**$\therefore$  Answer:  $L = e^{-2}$**

*Alternative check:  $\ln(e^x + 1) = x + \ln(1 + e^{-x})$ , so the exponent tends to  $-2$ .*

# FINAL ANSWERS + QUICK RULES

## Answer Sheet

| Q  | Final answer  |
|----|---------------|
| 1  | $3/2$         |
| 2  | $1/30$        |
| 3  | $\sqrt{2}/32$ |
| 4  | $n a^{n-1}$   |
| 5  | 6             |
| 6  | $1/2$         |
| 7  | $7a^3$        |
| 8  | $1/2$         |
| 9  | 0             |
| 10 | $1/2$         |
| 11 | 1             |
| 12 | $e^{-2}$      |

## QUICK RULES

- $0/0$  or  $\infty/\infty \rightarrow$  L'Hôpital is allowed.
- $0 \cdot \infty \rightarrow$  convert to a quotient.
- $\infty - \infty \rightarrow$  combine into one fraction/expression.
- $0^0, 1^\infty, \infty^0 \rightarrow$  take  $\ln$ , solve the exponent limit, then use  $e^{\text{answer}}$ .
- Prefer factorization, rationalization, and standard trig limits when they are shorter.

### Must-know standard limits:

$$\sin x/x \rightarrow 1 \quad (1-\cos x)/x^2 \rightarrow 1/2 \quad \ln(1+x)/x \rightarrow 1$$

$$x \ln x \rightarrow 0 \text{ as } x \rightarrow 0^+$$